PROJECT TIMELINES AND BASELINE SCHEDULING

MBA 666

Green Hydrogen Production Plant

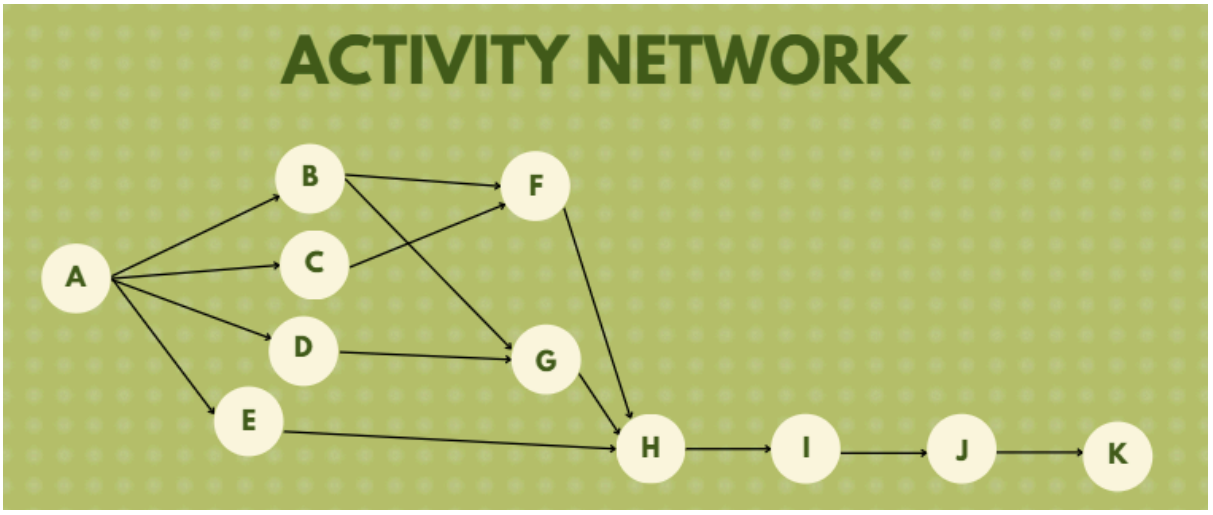
Submitted by:
Shrimi Agrawal(221036)

A. Project Baseline Schedule (CPM Analysis)

I. Activity List with CPM Attributes

ID	Activity	Predecessors	Duration (Week)
A	Initial Planning & Approvals	-	4
B	Site preparation & foundations	A	6
C	Power Input Procurement	A	12
D	Water System Procurement	A	8
E	Electrolyzer Procurement	A	14
F	Install Power Infrastructure(PV/BESS)	C,B	10
G	Install Water Systems	D,B	4
H	Install elec. & H2 Storage(PESO)	E,F,G	12
I	System Integration & SCADA	H	4
J	Testing	I	2

K	Closure(Regulatory sign-off)	J	2



II. Scheduling Flexibility Calculation using Slack

ID	Duration	ES(Wk)	EF(Wk)	LS(Wk)	LF(Wk)	Slack
A	4	0	4	0	4	0
B	6	4	10	10	16	6
C	12	4	16	4	16	0
D	8	4	12	14	22	10
E	14	4	18	12	26	8
F	10	16	26	16	26	0
G	4	12	16	22	26	10
H	12	26	38	26	38	0
I	4	38	42	38	42	0
J	2	42	44	42	44	0
K	2	44	46	44	46	0

ES: Early Start
EF: Early Finish (EF = ES + D)
D: Duration of activity
LS:Late Start
LF:Late Finish(LF = LS + D)

Slack = LF-EF

III. Critical Path Identification

Path	Activity Sequence	Length(Weeks)	Total Duration
Path 1	A->C->F->H->I->J->K	4+12+10+12+4+2+2	46
Path 2	A->B->F->H->I->J->K	4+6+10+12+4+2+2	40
Path 3	A->B->G->H->I->J->K	4+6+4+12+4+2+2	34
Path 4	A->D->G->H->I->J->K	4+8+4+12+4+2+2	36
Path 5	A->E->H->I->J->K	4+14+12+4+2+2	38

The critical path is **A->C->F->H->I->J->K** and the baseline project duration is **46 weeks**

B. Scheduling Risk Analysis (PERT)

We perform PERT analysis to add layer of risk assessment to the determined CPM schedule.
Based on assumptions about the nature of activities we assume value for To,Tp for the activities in the critical path

I.Rationale for PERT Time Estimates

CP Activity	To (Week)	Tm (Week)	Tp (Week)	Rationale for To and Tp value chosen by activity
A(Planning)	3	4	5	Low variability
C(Power Proc.)	10	12	18	High risk for logistics
F(Install Power)	8	10	14	Medium risk from weather
H(Install H2)	10	12	20	High risk from technical complexity
I(SCADA/ integration)	3	4	5	Low risk,short duration
J(Testing)	1	2	4	Low duration,high risk outcome
K(Sign-off)	1	2	3	Low variability

III. PERT calculation and Project Risk

ID	To (Wk)	Tm (Wk)	Tp (Wk)	Te (Wk)	σ^2 (Variance)
A	3	4	5	4.00	0.11
C	10	12	18	12.67	1.78
F	8	10	14	10.33	1.00
H	10	12	20	13.00	2.78
I	3	4	5	4.00	0.11
J	1	2	4	2.17	0.25
K	1	2	3	2.00	0.11
TOTAL				48.17	6.14

To : Optimist Duration
Tp: Pessimist Duration
Tm: CPM Duration
Te: Expected Project Duration
Te = (To + 4Tm + Tp) / 6
σ^2: Variance
σ^2 = $((Tp - To) / 6)^2$

The most likely completion time is **48.17 weeks** (two weeks longer than the deterministic 46-week estimate).
From above variance we get the standard deviation to be around $\sqrt{6.14} \approx$ 2.48 weeks.

On basis of PERT analysis we have result that project may take more that 46 weeks as decided by the critical path calculations.
To ensure that the project completes in decided time we look into crashing opportunities:

C. Cost-Time Optimization /Project Crashing

I. Financial Parameters

To determine the optimal duration, we calculate the costs from previous documents:

1. **Indirect Cost Saved(IC)**: Project overhead cost saved per week

IC saved = Annual OPEX/52 weeks
= ₹2,32,40,000/52
≈ **₹ 4,46,923 /week** [based on consideration that source of the Indirect Cost, is spread evenly across a full calendar year.]
We set this value as threshold to see if crashing is economical.

2. **Direct cost(DC) Slope Calculation**: Cost of acceleration calculated for each critical path activity

$$\text{DC Slope} = \frac{\text{Crash Cost} - \text{Normal Cost}}{\text{Normal Duration} - \text{Crash Duration}}$$

II. DC Slope Calculation and Prioritization

ID	D_normal (Wk)	D_crash (Wk)	Cost_normal (₹)	Cost_crash (₹)	DC_slope (₹/Wk)
A	4	3	10,00,000	14,00,000	4,00,000
B	6	4	20,00,000	25,00,000	2,50,000
C	12	10	60,00,000	63,00,000	1,50,000
D	8	6	15,00,000	20,00,000	2,50,000
E	14	12	70,00,000	78,00,000	4,00,000
F	10	8	40,00,000	45,00,000	2,50,000
G	4	3	8,00,000	11,00,000	3,00,000
H	12	9	50,00,000	62,00,000	4,00,000
I	4	3	5,00,000	10,00,000	5,00,000
J	2	1	3,00,000	5,00,000	2,00,000
K	2	1	2,00,000	5,00,000	3,00,000

Here we see activities **A ,C ,F ,H ,J ,K**are economically feasible and profitable to crash.

We don't crash B,D,E,G activities because of the positive slack calculated earlier.Crashing them only reduces activity's slack not the project's slack.

Now we crash the **A ,C ,F ,H ,J ,K** activities to get optimised duration:

1. Crash C by 2 weeks:
   New DC = 2 x 1,50,000 = ₹ 3,00,000
   New duration = 46-2 = 44 week
   Net savings =₹ 4,46,923 x 2 - ₹ 3,00,000 = ₹ 5,93,846
2. Crash J by 1 week:
   New DC = 1 x 2,00,000 = ₹ 2,00,000
   New duration = 44-1 = 43 week
   Net savings =₹4,46,923 - ₹2,00,000 = ₹ 2,46,923
3. Crash F by 2 weeks:
   New DC = 2 x 2,50,000 = ₹ 5,00,000
   New duration = 43-2 = 41 week
   Net savings =₹4,46,923 x 2 - ₹5,00,000 = ₹ 3,93,846
4. Crash K by 1 week:
   New DC = 1 x 3,00,000 = ₹ 3,00,000
   New duration = 41-1 = 40 week
   Net savings =₹ 4,46,923 - ₹3,00,000= ₹1,46,923
5. Crash A by 1 week:
   New DC = 1 x 4,00,000 = ₹ 4,00,000
   New duration = 40-1 = 39 week
   Net savings = ₹4,46,923 - ₹4,00,000 = ₹ 46,923
6. Crash H by 3 weeks:
   New DC = 3 x 4,00,000 = ₹ 12,00,000
   New duration = 39-3 = 36 week
   Net savings =₹ 4,46,923 x 3 - ₹ 12,00,000 = ₹1,40,769

Based on this iterative process we get a total economic benefit from the accelerated schedule of **₹15,69,230.**
The total direct crashing cost is **₹29,00,000.**
Also we get the new optimal duration for **36 weeks**.